STEIN FACTORIZATION FOR PROPER MORPHISMS OF ALGEBRAIC SPACES: INTEGRAL TARGET WITH CONNECTED GEOMETRIC FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8196-8215 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8217-8281, the complete author proof (63 body lines).

Editorial note. Navigation stubs: 12 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Revived hidden-pool card: lane-s9 pair ruling counted this B side; pick2's earlier all-label veto had hidden it.

Editorial note. Cites the two same-wave cards L16209/L16203 — citation chain closes within wave 20.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas stein-universally-closed (L16209), characterize-geometrically-connected-fibres; theorems stein-factorization-Noetherian (L16203); Limits of Spaces descent lemmas; Morphisms/Cohomology of Spaces proper base-change inputs.

Source reference. Original theorem 03GZ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8099–8105. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 03GY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7994–8014. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 073K: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0827: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original theorem 07SI: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 07SQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0A0X: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0822: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

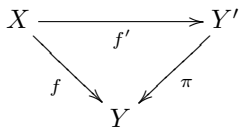
Source reference. Original theorem 04WP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 03KR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0462: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8281–8281. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 03H0: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8127–8145. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1 (Stein factorization; general case)**.** *Let S be a scheme. Let $f : X \to Y$ be a proper morphism of algebraic spaces over S. There exists a factorization*



with the following properties:

1. *the morphism f' is proper with connected geometric fibres,*
2. *the morphism $\pi : Y' \to Y$ is integral,*
3. *we have $f'_*\mathcal{O}_X = \mathcal{O}_{Y'}$,*
4. *we have $Y' = \underline{\mathrm{Spec}}_Y(f_*\mathcal{O}_X)$, and*
5. *Y' is the normalization of Y in X (Morphisms of Spaces, Definition 0822).*

Proof. We may apply Lemma 03GY to get the morphism $f' : X \to Y'$. Note that besides the conclusions of Lemma 03GY we also have that f' is separated (Morphisms of Spaces, Lemma 03KR) and finite type (Morphisms of Spaces, Lemma

0462). Hence f' is proper. At this point we have proved all of the statements except for the statement that f' has connected geometric fibres.

It is clear from the discussion that we may replace Y by Y'. Then $f : X \to Y$ is proper and $f_*\mathcal{O}_X = \mathcal{O}_Y$. Note that these conditions are preserved under flat base change (Morphisms of Spaces, Lemma 04WP and Cohomology of Spaces, Lemma 073K). Let $\overline{y}$ be a geometric point of Y. By Lemma 03GZ and the remark just made we reduce to the case where Y is a scheme, $y \in Y$ is a point, $f : X \to Y$ is a proper algebraic space over Y with $f_*\mathcal{O}_X = \mathcal{O}_Y$, and we have to show the fibre X_y is connected. Replacing Y by an affine neighbourhood of y we may assume that $Y = \mathrm{Spec}(R)$ is affine. Then $f_*\mathcal{O}_X = \mathcal{O}_Y$ signifies that the ring map $R \to \Gamma(X, \mathcal{O}_X)$ is bijective.

By Limits of Spaces, Lemma 0A0X we can write $(X \to Y) = \lim(X_i \to Y_i)$ with $X_i \to Y_i$ proper and of finite presentation and Y_i Noetherian. For i large enough Y_i is affine (Limits of Spaces, Lemma 07SQ). Say $Y_i = \mathrm{Spec}(R_i)$. Let $R_i' = \Gamma(X_i, \mathcal{O}_{X_i})$. Observe that we have ring maps $R_i \to R_i' \to R$. Namely, we have the first because X_i is an algebraic space over R_i and the second because we have $X \to X_i$ and $R = \Gamma(X, \mathcal{O}_X)$. Note that $R = \mathrm{colim}\, R_i'$ by Limits of Spaces, Lemma 07SI. Then

$$\begin{array}{ccccc} X & \longrightarrow & X_i & & \\ \downarrow & & \downarrow & & \\ Y & \longrightarrow & Y_i' & \longrightarrow & Y_i \end{array}$$

is commutative with $Y_i' = \mathrm{Spec}(R_i')$. Let $y_i' \in Y_i'$ be the image of y. We have $X_y = \lim X_{i,y_i'}$ because $X = \lim X_i$, $Y = \lim Y_i'$, and $\kappa(y) = \mathrm{colim}\, \kappa(y_i')$. Now let $X_y = U \amalg V$ with U and V open and closed. Then U, V are the inverse images of opens U_i, V_i in $X_{i,y_i'}$ (Limits of Spaces, Lemma 0827). By Theorem 03H0 the fibres of $X_i \to Y_i'$ are connected, hence either U or V is empty. This finishes the proof. $\square$